INVERSE DYNAMICS CONTROL.

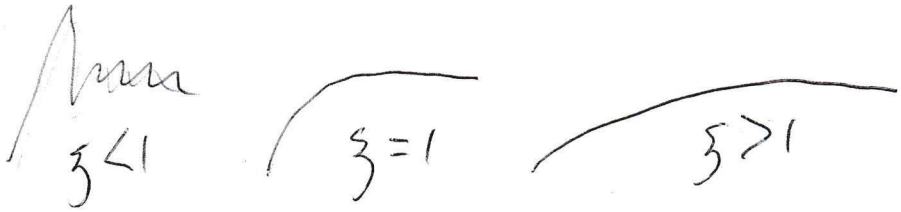
ACTUAL RESPONSE = DESIRED RESPONSE

FREQUENCY DOMAIN

$$(s^2 + 2\xi\omega_n s + \omega_n^2)Q(s) = (s^2 + 2\xi\omega_n s + \omega_n^2)Q_D(s)$$

TIME DOMAIN

$$\ddot{q} + 2\xi\omega_n\dot{q} + \omega_n^2 q = \ddot{q}_D + 2\xi\omega_n\dot{q}_D + \omega_n^2 q_D.$$



REARRANGE OUR EQUATION:

$$\ddot{q} = \ddot{q}_D + 2\xi\omega_n(\dot{q}_D - \dot{q}) + \omega_n^2(q_D - q).$$

LET $y = \ddot{q}$

$$\therefore \quad y = \ddot{q}_D + 2\xi\omega_n(\dot{q}_D - \dot{q}) + \omega_n^2(q_D - q). \quad -①$$

USE y IN OUR DYNAMIC EQUATION IN ORDER TO REQUIRED INPUT COMMANDS FOR THE ROBOT.

$$u = M(q)\,y + N(\dot{q}, q) \quad -②$$

SUBSTITUTE ① INTO ②

$$u = M(q)\left(\ddot{q}_D + 2\xi\omega_n(\dot{q}_D - \dot{q}) + \omega_n^2(q_D - q)\right) + N(\dot{q}, q)$$

LET $G_D = \mathrm{diag}(2\xi\omega_n)$ AND $G_P = \mathrm{diag}(\omega_n^2)$

$$\therefore \quad u = M(q)\left(\ddot{q}_D + G_D\Delta\dot{q} + G_P\Delta q\right) + N(\dot{q}, q)$$